

PilotFish RESTful Web Service Transport – Complete Reference (26R1.11 WAR)

Derived from decompiled eip.war.hs.26R1.11

August 4, 2026

At a glance (plain English)

This is a **REST client** for the end of a route. Configure a resource URL, HTTP method (GET/POST/PUT/DELETE/HEAD), media type, headers and query parameters – then the transport calls the API (Jakarta JAX-RS client) and puts the **response bytes** into the transaction.

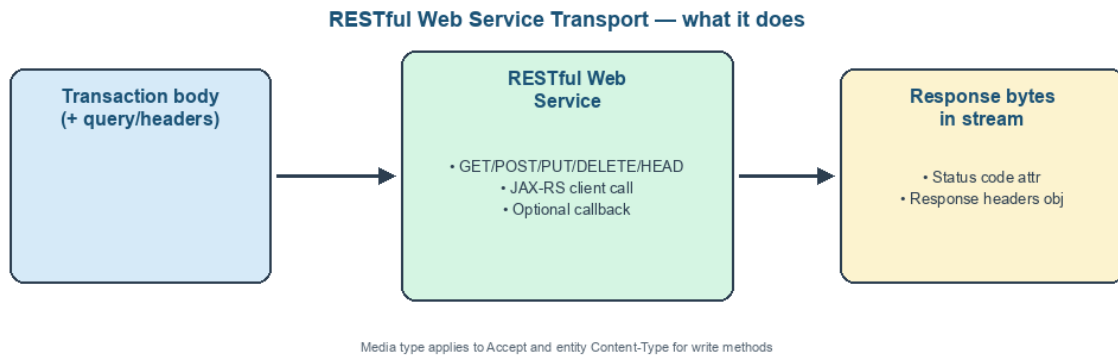


Figure 1: At a glance (plain English)

- **Calls** RESTful endpoints with method + Accept/entity media type.
- **Merges** configured headers with StoredHeaders attribute map.
- **Records** HTTP status and response headers on the transaction.
- **Optionally** triggers a programmable Response Listener (including empty responses).

Purpose

Deep dive into the **RESTful Web Service** transport in **eip.war.hs.26R1.11**: JAX-RS invocation paths, settings object, SSLAuthUtil, callback behavior, and attributes.

Provenance	Value
WAR	eip.war.hs.26R1.11
Module JAR	xcs-modules-http-1.0.3.jar
Decompiler	CFR 0.152
Implementation class	com.pilotfish.eip.modules.http.rest.RESTfu
Decompiled path	war-pipeline/decompiled/eip-26R1.11/.../RE
Transport type string	RESTful Web Service
Description	This Transport allows to perform calls to the RESTful-style web services
Help ID	console.transport.RESTfulWebService
Categories	WEB, WEB_SERVICE
Client API	Jakarta ClientBuilder / Invocation.Builder

1. Conceptual model

```
Build RESTfulWebServiceTransportSettings
|
v
ClientBuilder (SSL, hostname, timeouts)
|
v
target(resourcePath) + query params
request(mediaType) + headers
|
v
GET | POST(entity stream) | PUT | DELETE | HEAD
|
v
status + response headers attrs
response body bytes -> data stream
[optional] trigger Response Listener
```

2. High-level behavior

2.1 Outer `executeTransport(TransactionData)`

1. Populate settings: resource path, timeout, media type, method, username/password, query params table (parsed).
2. `byte[] response = executeTransport(data, settings).`
3. If Response Listener name non-empty AND (response length > 0 OR Call on empty response): `TriggerableListenerUtils.trigger.`
4. `data.setDataStream(new ByteArrayInputStream(response)).`

Wrapped by `AbstractTransport sendData` (complete/error).

2.2 Inner `executeTransport(data, settings)`

1. If username and password both non-empty: `Authenticator.setDefault` with `UsernamePassAuthenticator` (**JVM-global** default authenticator).
2. `SSLContext` / `HostnameVerifier` from `SSLAuthUtil`.
3. Connect + read timeouts = settings timeout **seconds**.
4. Build `WebTarget`; apply query param pairs.
5. `MediaType.valueOf(mediaType)` – invalid throws.
6. Consolidate headers: config table then `StoredHeaders` map (table first, attribute overlay second).
7. Dispatch method:
 - GET / DELETE / HEAD – no entity
 - POST / PUT – `Entity.entity(dataStream, mediaType)`
8. Collect response headers into `com.pilotfish.http.ResponseHeaders` (`HttpResponse` object).
9. Set `com.pilotfish.HTTPResponseCode` as string.
10. Read entity bytes (empty array if response null).

No HTTP status-based failure in this class – non-2xx still returns body and status attribute.

3. Configuration reference

3.1 Module-specific options

UI label	Tag	Type	Default	Tab	Required	Nuances
Resource Path	<code>RESOURCE_PATH</code>	String	""	Basic	Yes	Full URI
Accept Media Type	<code>ACCEPT_MEDIA_TYPE</code>	String	*/*	Basic	Yes	Also entity type for POST/PUT
Method	<code>METHOD_TO_EXECUTE</code>	Enum	GET	Basic	No	GET, POST, PUT, DELETE, HEAD

UI label	Tag	Type	Default	Tab	Required	Nuances
Connection/Read Timeout	TIMEOUT	Integer	0	Basic	Yes	Seconds; 0 = no timeout
HTTP Headers	HTTP_HEADERS	Table	empty	HTTP Headers	No	Key/Value
Query Parameters	QUERY_PARAMS	Table	empty	Basic	No	Parsed with transaction tokens
User Name	USER_NAME	String	""	Authentication	No	With password installs JVM Authenticator
Password	PASSWORD	Password	""	Authentication	No	
Response Listener	CALLBACK	Listener	""	Response	No	Programmatic listener
Call response listener on empty response	CALL_ON_EMPTY_RESPONSE	Boolean	FALSE	Response	No	

3.2 SSL (SSLAuthUtil.getSSLConfigItems, all tabs)

Same structural set as HTTP Post SSL section: TrustSelfSignedCerts, ValidateHostname, SpecifyTLS/TLSVersion, KeyStore, *TrustStoreSource*, CertificateAttribute, etc. `readConfiguration` applies `adjustForBackwardsCompatibility`.

3.3 Inherited AbstractTransport

Standard `sendData -> checkRequiredData -> executeTransport -> complete`. Implements `AbstractReferencedListeners` for CALLBACK rename (`referencedProgramListenerNameIsUsed` uses raw CALLBACK value).

4. Transaction attributes

4.1 Written

Attribute key	Type / content
<code>com.pilotfish.HTTPResponseCode</code>	Status as string
<code>com.pilotfish.http.ResponseHeaders</code>	<code>HttpResponse</code> aggregator of response headers

4.2 Consumed

Attribute key	Usage
<code>com.pilotfish.eip.modules.http.StoredHeaders</code>	Must be a <code>Map</code> ; else <code>EIPEException</code>

5. Worked example – POST JSON with callback

- Resource Path: `https://api.example.com/v1/claims`
- Method: `POST`
- Accept Media Type: `application/json`
- Timeout: `30`
- Response Listener: `Process-Rest-Reply`
- Call on empty: `false`

Transaction JSON body is POSTed. Response JSON replaces stream and triggers listener when non-empty. Check `com.pilotfish.HTTPResponseCode` for 201/400 even though transport did not throw.

6. Known quirks and caveats

1. **No status throwing** – unlike HTTP Post Default mode.
2. **JVM Authenticator.setDefault** leaks across other HTTP work on the JVM until replaced.
3. **Same media type** for Accept and entity Content-Type on write methods.
4. **HEAD/GET ignore** transaction body for the request entity.
5. **Empty GET response** still replaces stream with empty bytes.
6. **Non-Map StoredHeaders** fails hard.

7. Operational recommendations

Goal	Guidance
Treat 4xx as fail	Add processor checking <code>com.pilotfish.HTTPResponseCode</code> or use HTTP Post with code handling
Auth	Prefer Authorization header over JVM Authenticator when possible
Timeouts	Set non-zero Timeout for production
Callbacks	Enable Call on empty only when empty body is still a meaningful signal

8. Source index (this WAR)

Topic	Path under war-pipeline/decompiled/eip-26R1.11/
RESTful transport	com/pilotfish/eip/modules/http/rest/RESTfulWebServiceTransport.java
Settings	com/pilotfish/eip/modules/http/rest/RESTfulWebServiceTransportSettings.java
SSL util	com/pilotfish/eip/modules/http/SSLAuthUtil.java
Abstract transport	com/pilotfish/eip/extend/AbstractTransport.java
Module JAR	war-pipeline/jars/eip-26R1.11/xcs-modules-http-1.0.3.jar

9. Pipeline provenance

PilotFish WARs/eip.war.hs.26R1.11

-> xcs-modules-http-1.0.3.jar

-> CFR decompile

-> Documents/Transports/26R1.11/PilotFish-RESTful-Web-Service-Transport-Reference-26R1.11.(m

Deep-dive documentation from WAR decompile – RESTful Web Service Transport – 26R1.11 – August 2026.